CV

Quiz navigation

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Question **1**
Not yet answered
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Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$\hat{y} = w_1 x + w_0$.

Write the batch gradient descent update for w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}$, where the learning rate is η. Numerical application: $(1,1), (2,-1), (3,2)$, $\eta = 0.1$.

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Question **2**
Not yet answered
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Assume we have a training set for two-class classification in two dimensions that contains seven sample points: points $(1,1), (2,2), (3,1)$ with label $+1$ and points $(4,5), (5,7), (6,5), (7,7)$ with label -1. Which points are the support vectors for a hard-margin SVM?

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Question **2**
Not yet answered
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Assume we have a training set for two-class classification in two dimensions that contains seven sample points: points $(1,1),(2,2),(3,1)$ with label $+1$ and points $(4,5),(5,7),(6,5),(7,7)$ with label -1. Which points are the support vectors for a hard-margin SVM?

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Question **3**
Not yet answered
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Consider training a classification decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of X, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest Gini index: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

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Question **3**

Not yet answered

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Consider training a classification decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of X, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest Gini index: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

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Question **4**

Not yet answered

Assume we have the design matrix

$$\begin{bmatrix} 6 & -4 \end{bmatrix}$$

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Question **4**
Not yet answered
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Flag question

Assume we have the design matrix

$$X = \begin{bmatrix} 6 & -4 \\ -3 & 5 \\ -2 & 6 \\ 7 & -3 \end{bmatrix}.$$

Use PCA to reduce the dimension from 2 to 1, i.e., to compute the first principal component. Observe that X is not centered.

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Question **5**
Not yet answered
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Consider a dataset containing six one-dimensional points: $\{2, 4, 7, 8, 12, 14\}$. After three iterations of hierarchical clustering using Euclidean distance between points, we get the 3 clusters: $C_1 = \{2, 4\}$, $C_2 = \{7, 8\}$, and $C_3 = \{12, 14\}$. What is the distance between clusters C_1 and C_2 using single linkage? What is the distance between clusters C_1 and C_2 using complete linkage?

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Question **5**

Not yet answered

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Flag question

Consider a dataset containing six one-dimensional points: $\{2, 4, 7, 8, 12, 14\}$. After three iterations of hierarchical clustering using Euclidean distance between points, we get the 3 clusters: $C_1 = \{2, 4\}$, $C_2 = \{7, 8\}$, and $C_3 = \{12, 14\}$. What is the distance between clusters C_1 and C_2 using single linkage? What is the distance between clusters C_1 and C_2 using complete linkage?

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Question **6**

Not yet answered

The following graph shows the structure of a simple neural network with a single hidden layer. The input layer consists of three dimensions $x = (x_1, x_2, x_3)$. The hidden layer includes two units $h = (h_1, h_2)$. The output layer includes one unit y. We ignore bias terms for simplicity. We use sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ as activation function for the hidden layer and the identity for the output

CV

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Question **6**

Not yet answered

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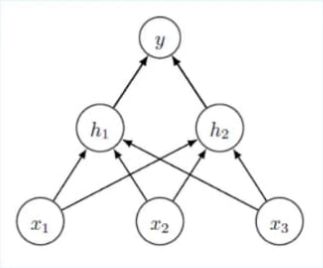
The following graph shows the structure of a simple neural network with a single hidden layer. The input layer consists of three dimensions $x = (x_1, x_2, x_3)$. The hidden layer includes two units $h = (h_1, h_2)$. The output layer includes one unit y. We ignore bias terms for simplicity. We use sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ as activation function for the hidden layer and the identity for the output layer. Moreover, denote by $l(y,t) = \frac{1}{2}(y-t)^2$ the loss function. Here, t is the target value for the output unit y. Denote by W and V weight matrices connecting input and hidden layer, and hidden layer and output, respectively. Write out symbolically the mapping y as a function of x using $\sigma(z), W, V$. Compute symbolically $\frac{\partial l}{\partial V}$. Numerical application:

$$W = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}, V = \begin{bmatrix} 0 & 1 \end{bmatrix}, x = (1, 2, 1), t = 1.$$



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Question **7**
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Assume we have an input image whose shape is $128 \times 128 \times 3$. Compute the output size and number of parameters of each of the following possible settings for the first convolutional layer, given the input has the shape described above.

(a) The first convolutional layer has a stride of 1, a filter size of 3, input padding of 0, and 64 filters.

(b) The first convolutional layer has a stride of 1, a filter size of 5, input padding of 2, and 16 filters.

(c) The first convolutional layer has a stride of 2, a filter size of 2, input padding of 0, and 32 filters.

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Question **8**
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Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

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Question **8**
Not yet answered
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Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

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Question **9**
Not yet answered
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Assume a scalar multiplication and a scalar addition both account for one FLOP and exponentiation accounts for 10 FLOPs. All other scalar operations account for 0 FLOPs. How many FLOPs does the scaled dot-product attention require for n queries and m key-value pairs, where queries and keys are of length d and values are of length v. Give the answer symbolically. Numerical application: $n = 2, m = 3, d = 2, v = 3$.

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Examen - Prezentarea 1

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Time left 1:38:13

Question **1**

Not yet answered

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Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$\hat{y} = w_1 x + w_0.$

Compute w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}$. Numerical application: $(1, 1), (2, -1), (3, 2)$.

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Examen - Prezentarea 1

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Question **9**

Not yet answered

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Compute the total number of parameters for additive attention for n queries and m key-value pairs, where queries are of length q, keys are of length k, and values are of length v. Give the answer symbolically. Numerical application:
$n = 2, m = 3, q = k = 2, v = 3.$

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Question **10**

Not yet answered

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Examen - Prezentarea 1

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Question **7**

Not yet answered

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Assume we have the following CNN classifier:
INPUT, CONV-9-32, POOL-2, CONV-5-64, POOL-2, CONV-5-64, POOL-2, FC-3.
For each layer, calculate the number of weights, number of biases, and the size of the associated feature maps. The size of INPUT is $128 \times 128 \times 3$. The notation follows the convention:

- CONV-K-N denotes a convolutional layer with N filters, each of them of size K×K, padding and stride parameters are always 0 and 1, respectively.
- POOL-K indicates a K×K pooling layer with stride K and padding 0.
- FC-N stands for a fully-connected layer with N neurons.

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Examen - Prezentarea 1

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Question **3**

Not yet answered

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Flag question

Consider training a classification decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of X, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest entropy: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

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